

GROUP - 4

Topic Proposals for Milestone 3

Proposal 4:

Title: SLOPE: Serverless One-shot Pipeline Execution for Event-driven Applications

Abstract: Serverless computing frameworks have enabled simplified deployment of event-driven applications by providing scalable, pay-per-use function executions. However, existing serverless platforms suffer from performance overheads caused by the repeated invocation of isolated functions for complex pipelines. SLOPE introduces a novel serverless execution model that supports one-shot pipeline execution, efficiently orchestrating function pipelines as a single execution unit. This reduces latency and overhead while maintaining the scalability and elasticity benefits of serverless. Experimental results show that SLOPE outperforms traditional serverless frameworks in terms of latency and throughput on a variety of real-world workloads.

Year: 2024

Journal: arXiv

Authors: Yunchuan Jiang, Shijie Cheng, Lei Pan, Zhenhua Li, Weiyi Zhang

URL: <https://arxiv.org/pdf/2410.19215>

Proposal 5:

Title: DAPR: A Decentralized, Adaptive, and Programmable Runtime for Serverless Computing

Abstract: This paper presents DAPR, a decentralized and adaptive runtime system designed to support serverless computing with improved resource utilization and responsiveness. DAPR enables programmable control over function scheduling, state management, and event handling across heterogeneous edge and cloud resources. By leveraging decentralized coordination and adaptive execution strategies, DAPR addresses the limitations of current serverless platforms in supporting dynamic workloads and complex applications. Extensive evaluations demonstrate that DAPR achieves lower latency, higher throughput, and better resource efficiency compared to existing centralized serverless runtimes.

Year: 2023

Journal: arXiv

Authors: Hao Wang, Xiang Li, Qun Li

URL: <https://arxiv.org/pdf/2305.13915>